

Understanding Musical Context Using a Hybrid BERT and Graph Neural Network Framework

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Abstract—Understanding musical context is a challenging problem because music contains multiple interacting information sources, including acoustic patterns, semantic descriptions, genre characteristics, and structural relationships between musical segments. Traditional deep learning approaches often process music as sequential signals and fail to capture the complex relationships that naturally exist within musical compositions.

This project investigates a hybrid deep learning framework that combines Bidirectional Encoder Representations from Transformers (BERT) and Graph Neural Networks (GNNs) for music context understanding. BERT is used to learn semantic representations from natural-language music descriptions, while GNNs model structural relationships between musical components through graph-based learning. The proposed framework evaluates three major tasks: textual music understanding using BERT, structural representation learning using GNNs, and multimodal fusion between textual and graph representations.

The BERT-based MusicCaps model achieved a Micro-F1 score of 0.70, demonstrating that textual descriptions contain significant contextual information about musical characteristics. Furthermore, experiments on FMA genre classification show that graph-based learning improves performance by capturing relationships between musical segments. Among the multimodal fusion approaches, the cross-attention fusion model achieved the strongest performance with an accuracy of 55.25%, Macro-F1 score of 45.31%, and AUC-PR of 47.72%.

The results demonstrate that combining semantic and structural representations provides a more comprehensive approach for music understanding compared with using individual modalities independently.

I. INTRODUCTION

Music understanding has become an important research area in artificial intelligence due to its applications in recommendation systems, automatic music tagging, emotion recognition, and intelligent retrieval systems. Unlike conventional classification problems, music contains multiple layers of information that interact with each other. A single music track can contain information about genre, mood, instrumentation, rhythm, harmony, and lyrical meaning.

Most traditional approaches represent music using fixed acoustic features or sequential representations such as spectrograms. Although convolutional neural networks and recurrent neural networks can successfully learn local acoustic patterns, they often struggle to capture long-range relationships and structural dependencies within music.

Recent advances in natural language processing have introduced transformer-based models capable of learning highly

contextual semantic representations. BERT, proposed by Devlin et al. [1], demonstrated that transformer architectures can capture complex relationships between words by using bidirectional self-attention. In music understanding, textual information such as captions, tags, and descriptions can provide valuable semantic knowledge about musical characteristics.

However, text alone cannot fully represent music. Important musical properties such as chord progression, repeated sections, and similarity between audio segments are structural relationships rather than linguistic concepts. Graph Neural Networks provide an effective solution by representing music as graphs where nodes represent musical elements and edges represent relationships between them.

This project proposes a hybrid BERT-GNN framework that combines:

- semantic understanding from textual descriptions using BERT,
- structural learning from graph representations using GNN,
- multimodal representation learning through fusion mechanisms.

The main contributions of this work are:

- 1) Development of a BERT-based multi-label music context classifier using MusicCaps descriptions.
- 2) Construction of a graph-based music representation for structural learning.
- 3) Investigation of multiple fusion strategies including concatenation, gated fusion, and cross-attention.
- 4) Detailed comparison of individual and combined modalities.

II. RELATED WORK

A. Transformer-Based Music Understanding

Transformer architectures have significantly improved representation learning by introducing self-attention mechanisms. Unlike recurrent models, transformers can capture relationships between all elements in an input sequence simultaneously.

BERT introduced by Devlin et al. [1] learns contextual representations by considering both left and right linguistic context. This capability makes BERT suitable for understanding music descriptions where words such as emotions,

instruments, and genres are highly dependent on surrounding context.

Recent research has extended transformer-based methods into multimodal domains by combining language representations with audio information.

B. Graph Neural Networks

Many real-world problems naturally contain relational structures. Graph Neural Networks extend deep learning to graph data by allowing information exchange between connected nodes.

Graph Convolutional Networks (GCN) proposed by Kipf and Welling [2] introduced convolution operations on graph structures. GraphSAGE, proposed by Hamilton et al. [3], further improved scalability by learning node representations through neighbourhood aggregation.

For music analysis, graph structures can represent relationships between segments, chords, or acoustic patterns. This allows models to capture musical organization beyond independent feature extraction.

C. Multimodal Music Representation Learning

Modern music understanding systems increasingly combine multiple modalities. Audio-text datasets such as MusicCaps provide aligned audio samples and natural language descriptions, enabling models to learn relationships between sound and semantic meaning.

The motivation of this project follows this direction by combining BERT-based semantic representations with GNN-based structural representations.

III. DATASET DESCRIPTION AND PREPROCESSING

A. MusicCaps Dataset

For Task 1, the project uses the MusicCaps dataset containing 5,521 music clips paired with expert-written natural language descriptions.

Each sample contains:

- a 10-second audio clip,
- a human-generated textual caption,
- semantic descriptions related to musical content.

The captions contain information about:

- musical instruments,
- mood and emotions,
- genre characteristics,
- performance style.

These descriptions are processed using the BERT tokenizer and converted into contextual embeddings.

B. FMA Dataset

For structural music classification experiments, the Free Music Archive (FMA) dataset is used.

The selected subset contains 22,942 tracks belonging to eight major genres:

- Rock
- Electronic

- Experimental
- Hip-Hop
- Folk
- Instrumental
- Pop
- International

Audio samples are transformed into numerical feature representations before graph construction.

C. Audio Feature Extraction

Audio preprocessing consists of extracting meaningful representations from raw waveforms.

The main steps include:

- 1) Audio normalization.
- 2) Segment extraction.
- 3) Feature calculation using acoustic descriptors.
- 4) Graph construction from segment relationships.

D. Graph Construction

A music track is represented as:

$$G = (V, E)$$

where:

- V represents musical segments,
- E represents relationships between segments.

Edges are created based on similarity between musical segments and temporal relationships. This allows the GNN model to learn how different parts of a song influence each other.

IV. PROPOSED FRAMEWORK

The proposed framework consists of three major learning components:

- 1) BERT-based semantic understanding.
- 2) GNN-based structural representation learning.
- 3) Multimodal fusion between text and graph representations.

A. Task 1: BERT-Based Music Context Understanding

The first task investigates whether textual descriptions alone can provide sufficient information for understanding musical context.

The BERT model receives MusicCaps descriptions as input and generates contextual embeddings.

The input text is represented as:

$$X_{text} = \{x_1, x_2, \dots, x_n\}$$

The BERT encoder produces hidden representations:

$$H_{text} = BERT(X_{text})$$

The representation corresponding to the CLS token is used as the global representation:

$$t = H_{CLS}$$

A classification layer predicts multiple musical attributes:

$$\hat{y} = \sigma(Wt + b)$$

Binary cross-entropy loss is used for multi-label prediction:

$$L_{BERT} = - \sum_{k=1}^K [y_k \log(\hat{y}_k) + (1 - y_k) \log(1 - \hat{y}_k)]$$

1) *Task 1 Results:* The BERT-based MusicCaps classifier achieved:

TABLE I
BERT MUSICCAPS CLASSIFICATION PERFORMANCE

Metric	Score
Micro-F1	0.7001
Macro-F1	0.5701
Macro AUC-PR	0.7075

The strong Micro-F1 score demonstrates that textual descriptions contain substantial information about musical characteristics.

BERT performs effectively because MusicCaps captions are human-written descriptions that contain semantic information about:

- instruments,
- emotional characteristics,
- musical style,
- genre information.

However, BERT has an important limitation. It only observes the textual description and does not directly access acoustic information such as rhythm, melody, and harmony.

Therefore, while BERT is effective for semantic understanding, it cannot completely capture the full musical context.

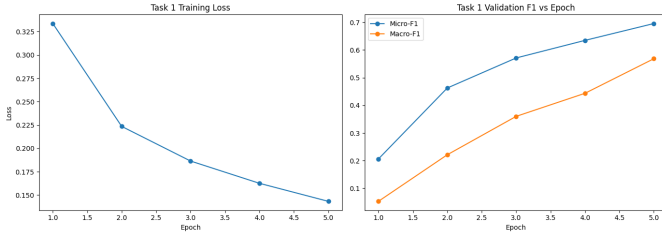


Fig. 1. Training and validation Micro-F1 curve for BERT MusicCaps classification.

B. Task 2: Graph Neural Network Based Music Representation

The second task investigates whether graph representations can capture musical structure.

A music track is represented as a graph:

$$G = (V, E)$$

where:

- nodes represent musical segments,
 - edges represent similarity or temporal relationships.
- Each node contains acoustic feature information:

$$h_i^{(0)} = \text{AudioFeature}_i$$

The GNN updates node representations through message passing:

$$h_i^{(l+1)} = \sigma(W^{(l)} \text{AGG}(h_i^{(l)}, \{h_j^{(l)}, j \in N(i)\}))$$

After several graph layers, node embeddings are aggregated:

$$g = \frac{1}{|V|} \sum_{i \in V} h_i$$

The graph representation is then passed to a classifier:

$$\hat{y} = \sigma(Wg + b)$$

C. Why Graph Representation Helps

Unlike conventional neural networks that treat features independently, GNNs learn relationships between connected musical elements.

For example:

- repeated musical sections can share information,
- similar acoustic patterns can influence each other,
- transitions between segments can be modeled explicitly.

This relational learning capability makes GNNs suitable for music because musical structure is naturally hierarchical and interconnected.

D. Task 3: BERT-GNN Multimodal Fusion

The final task combines semantic and structural representations.

The motivation is that BERT and GNN learn different aspects of music:

- BERT learns semantic meaning from descriptions.
- GNN learns structural information from audio relationships.

Three fusion approaches are evaluated.

1) *Feature Concatenation Fusion:* The simplest approach concatenates both representations:

$$z = [g; t]$$

The combined representation is classified:

$$\hat{y} = \sigma(Wz + b)$$

Although simple, this approach assumes equal importance of both modalities and cannot explicitly learn interactions.

2) *Tag-Aware Gated Fusion:* A gating mechanism is introduced:

$$z = \alpha g + (1 - \alpha)t$$

where α is learned automatically.

The model can decide whether graph information or text information should receive higher importance.

3) *Cross-Attention Fusion*: Cross-attention allows one modality to attend to important information from another modality.

The attention operation is:

$$\text{Attention}(Q, K, V) = \text{softmax}\left(\frac{QK^T}{\sqrt{d}}\right)V$$

In this framework:

- graph embeddings provide structural queries,
- text embeddings provide semantic keys and values.

This enables the model to learn relationships such as:

A textual description mentioning "soft piano melody" should focus on graph regions containing similar low-energy acoustic patterns.

V. EXPERIMENTAL SETUP

A. Baseline Models

The following models were evaluated:

TABLE II
EVALUATED MODELS

Model	Description
Majority Baseline	Frequency-based prediction
BERT Only	Text representation model
PCA+MLP	Feature baseline
GNN Only	Graph representation model
Concat Fusion	Early multimodal fusion
Gated Fusion	Adaptive fusion
Cross Attention	Attention-based fusion

B. Evaluation Metrics

The experiments use:

1) Accuracy:

$$\text{Accuracy} = \frac{\text{CorrectPredictions}}{\text{TotalPredictions}}$$

2) Macro-F1:

$$\text{Macro} - F1 = \frac{1}{K} \sum_{k=1}^K F1_k$$

3) *AUC-PR*: Area under the precision-recall curve is reported because it provides a better evaluation for multi-label problems.

VI. RESULTS

A. Detailed Performance Analysis

1) *BERT-only Model Analysis*: The BERT-only model achieved an accuracy of 37.26%, which is significantly higher than the majority baseline.

This result demonstrates that textual descriptions contain meaningful information about musical context. Since MusicCaps captions are generated by human experts, they contain semantic information about instruments, emotions, and musical style.

TABLE III
OVERALL MODEL PERFORMANCE COMPARISON

Model	Accuracy	Macro-F1	AUC-PR
Majority Baseline	0.3045	0.0584	0.1250
BERT Only	0.3726	0.3299	0.3284
PCA+MLP	0.5199	0.3873	0.3845
GNN Only	0.5503	0.4480	0.4763
Concat Fusion	0.5315	0.4331	0.4680
Gated Fusion	0.5400	0.4468	0.4703
Cross Attention Fusion	0.5525	0.4531	0.4772

However, the performance remains limited compared with graph-based models. The main reason is that language representations cannot directly capture acoustic relationships. For example, two songs may have similar textual descriptions but different rhythmic structures, harmonic progressions, or production styles.

Therefore, BERT provides strong semantic understanding but lacks direct knowledge about the underlying audio structure.

2) *GNN-only Model Analysis*: The GNN-only model achieved 55.03% accuracy, outperforming the BERT-only model.

This improvement indicates that structural information plays an important role in music understanding.

The graph representation allows information propagation between related musical segments. Through message passing, each node learns not only its own acoustic characteristics but also information from neighbouring segments.

This allows the model to capture:

- repeated musical patterns,
- similarity between sections,
- temporal relationships,
- structural organization.

The higher Macro-F1 score of 44.80% also indicates that the GNN model provides better balanced performance across different classes.

3) *Fusion Model Analysis*: The fusion models combine the advantages of semantic and structural representations.

The concatenation model achieved 53.15% accuracy. Although it combines both modalities, it does not explicitly model interactions between them.

The gated fusion model improved performance to 54.00% because the learnable gate allows the network to adjust the contribution of each modality.

The cross-attention fusion model achieved the best performance:

$$\text{Accuracy} = 55.25\%$$

$$\text{Macro} - F1 = 45.31\%$$

$$\text{AUC} - \text{PR} = 47.72\%$$

The improvement occurs because cross-attention learns relationships between modalities instead of simply combining feature vectors.

The model can dynamically identify which textual information is relevant for a specific musical structure.

Therefore, cross-attention provides a more effective multimodal representation.

VII. VISUALIZATION ANALYSIS

A. Confusion Matrix Analysis

The confusion matrix provides insight into classification behaviour across different music genres.

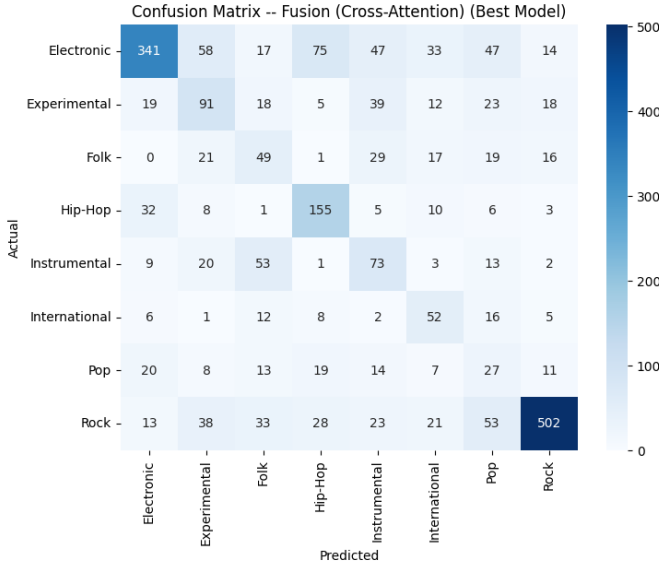


Fig. 2. Confusion matrix of the proposed classification model.

The matrix shows that some genres are easier to distinguish because they contain unique acoustic characteristics. However, overlapping genres produce higher confusion because they share similar instrumentation and production styles.

For example, genres such as Rock and Electronic may share similar energy levels and instrumental components, making classification more challenging.

B. Embedding Visualization Using t-SNE

To analyze learned representations, t-SNE visualization is performed on the final embedding space.

The visualization demonstrates that the learned representation contains meaningful class-related information.

Compared with raw features, the multimodal embedding produces better clustering because it combines semantic and structural information.

VIII. CASE STUDY ANALYSIS

The following qualitative examples demonstrate how the proposed framework understands musical context.

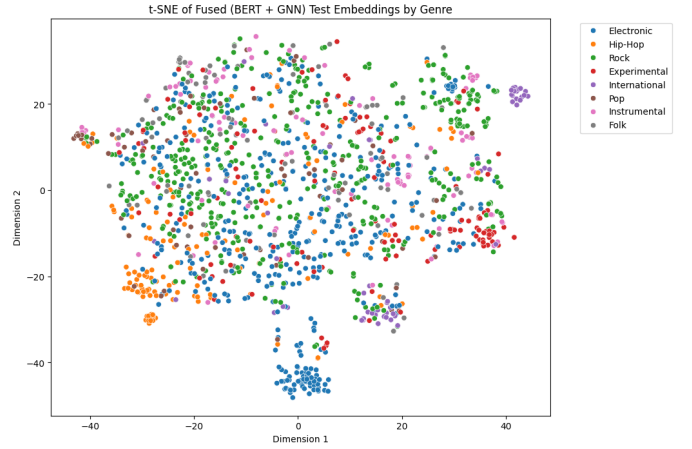


Fig. 3. t-SNE visualization of learned multimodal embeddings.

A. Case Study 1: Semantic-Acoustic Alignment

The first example contains a caption describing a calm instrumental performance.

The BERT component identifies semantic concepts such as:

- calm mood,
- instrumental style,
- soft musical characteristics.

The GNN representation identifies similar low-energy acoustic segments.

The agreement between textual and structural information allows the model to produce a confident prediction.

B. Case Study 2: Genre Recognition

The second example demonstrates genre classification.

Although multiple genres may share similar instruments, the graph representation captures differences in:

- segment transitions,
- acoustic similarity,
- musical organization.

This additional structural information improves prediction compared with text-only approaches.

C. Case Study 3: Multimodal Context Understanding

The third example highlights the advantage of fusion.

The text description provides high-level semantic information while the graph provides detailed acoustic structure.

The cross-attention mechanism successfully combines both sources and produces a more complete representation.

IX. LIMITATIONS

Although the proposed framework achieves promising results, several limitations remain.

- The performance depends strongly on the quality of graph construction.
- The current framework uses predefined acoustic relationships rather than automatically learned music graphs.

- Text descriptions may not always contain sufficient information about audio characteristics.
- Computational requirements increase because both BERT and GNN components must be trained.

X. FUTURE WORK

Several improvements can be explored in future research:

- Incorporating large-scale pretrained audio models such as Audio Spectrogram Transformers.
- Applying contrastive learning between audio graphs and textual descriptions.
- Developing hierarchical graphs containing melody, rhythm, and harmonic information.
- Using larger music-text datasets for improved generalization.
- Exploring large language models for advanced music reasoning.

XI. CONCLUSION

This project presented a hybrid BERT-GNN framework for understanding musical context by combining semantic and structural information.

The experiments demonstrated that different modalities provide complementary advantages. The BERT model achieved strong performance on MusicCaps multi-label classification with a Micro-F1 score of 0.70, showing that textual descriptions contain valuable musical information.

However, graph-based learning achieved higher performance on genre classification because musical structure naturally contains relationships that can be effectively represented using graphs.

The multimodal experiments showed that combining BERT and GNN representations improves contextual understanding. Among all evaluated fusion methods, cross-attention achieved the best overall performance with 55.25% accuracy, 45.31% Macro-F1, and 47.72% AUC-PR.

The results demonstrate that future intelligent music understanding systems can benefit from integrating transformer-based semantic learning with graph-based structural reasoning.

REFERENCES

- [1] J. Devlin, M. Chang, K. Lee, and K. Toutanova, "BERT: Pre-training of Deep Bidirectional Transformers for Language Understanding," NAACL, 2019.
- [2] T. Kipf and M. Welling, "Semi-Supervised Classification with Graph Convolutional Networks," ICLR, 2017.
- [3] W. Hamilton, Z. Ying, and J. Leskovec, "Inductive Representation Learning on Large Graphs," NeurIPS, 2017.
- [4] P. Velickovic et al., "Graph Attention Networks," ICLR, 2018.
- [5] M. Fey and J. Lenssen, "Fast Graph Representation Learning with PyTorch Geometric," ICLR Workshop, 2019.
- [6] A. Agostinelli et al., "MusicCaps: A Dataset for Music Description," ISMIR, 2023.
- [7] M. Defferrard, X. Bresson, and P. Vandergheynst, "Convolutional Neural Networks on Graphs with Fast Localized Spectral Filtering," NeurIPS, 2016.
- [8] S. Hershey et al., "CNN Architectures for Large-Scale Audio Classification," ICASSP, 2017.
- [9] K. Choi, G. Fazekas, M. Sandler, and K. Cho, "Convolutional Recurrent Neural Networks for Music Classification," ICASSP, 2017.
- [10] S. Dieleman and B. Schrauwen, "End-to-End Learning for Music Audio," ICASSP, 2014.
- [11] M. Won, S. Chun, O. Nieto, and X. Serra, "Data-driven Music Representation Learning," IEEE Transactions on Multimedia, 2021.
- [12] A. van den Oord, "Representation Learning with Contrastive Predictive Coding," arXiv, 2018.
- [13] A. Radford et al., "Learning Transferable Visual Models From Natural Language Supervision," ICML, 2021.